CBSE 2027 Chemistry — Predicted Paper (Set 4 of 10)

Based on the highly repeated questions from previous board papers (PYQs), official CBSE blueprints, and the latest NCERT syllabus for 2026-27, here is Set 4 of the predicted board paper. All questions are distinct from Sets 1, 2, and 3, maintaining a heavy focus on core numericals, conceptual reasoning, and crucial organic mechanisms.

SECTION A: Multiple Choice & Assertion-Reason Questions (1 Mark Each x 16 = 16 Marks)

Q1. The SI unit of molar conductivity (Λ_m) is: (a) $S \cdot m^{-1}$ (b) $S \cdot cm^2 \cdot mol^{-1}$ (c) $\Omega \cdot cm^{-1}$ (d) $S \cdot mol^{-1}$ * **Chapter:** Electrochemistry | **Confidence:** Very High

Q2. Two solutions having the same osmotic pressure at a given temperature are called: (a) Hypertonic solutions (b) Hypotonic solutions (c) Isotonic solutions (d) Azeotropic mixtures * **Chapter:** Solutions | **Confidence:** Very High

Q3. The rate constant for a reaction is $k = 3 \times 10^{-4} L \cdot mol^{-1} \cdot s^{-1}$. The overall order of the reaction is: (a) Zero order (b) First order (c) Second order (d) Third order * **Chapter:** Chemical Kinetics | **Confidence:** High

Q4. Which of the following lanthanoids is radioactive in nature? (a) Cerium (Ce) (b) Lutetium (Lu) (c) Promethium (Pm) (d) Gadolinium (Gd) * **Chapter:** d- and f-Block Elements | **Confidence:** Medium

Q5. What is the coordination number of Iron (Fe) in the complex ion $[Fe(C_2O_4)_3]^{3-}$? (a) 3 (b) 4 (c) 6 (d) 5 * **Chapter:** Coordination Compounds | **Confidence:** High

Q6. The main product of the reaction between a primary alkyl halide and silver nitrite ($AgNO_2$) is: (a) Alkyl nitrite (b) Nitroalkane (c) Alkyl nitrate (d) Silver alkoxide * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q7. Heating phenol with zinc dust yields which of the following organic compounds? (a) Benzene (b) Toluene (c) Benzoquinone (d) Zinc phenoxide * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q8. Which of the following compounds does **not** undergo an Aldol condensation reaction? (a) Acetaldehyde (b) Acetone (c) Trichloroacetaldehyde (d) Propanone * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

Q9. Which of the following is a tertiary (3°) amine? (a) tert-Butylamine (b) N-Methylaniline (c) N,N-Dimethylaniline (d) Isopropylamine * **Chapter:** Amines | **Confidence:** High

Q10. The specific sugar moiety present in DNA molecules is: (a) β-D-ribose (b) α-D-glucose (c) β-D-2-deoxyribose (d) α-D-fructose * **Chapter:** Biomolecules | **Confidence:** Very High

Q11. Which of the following alkyl halides is optically active? (a) 1-Chlorobutane (b) 2-Chlorobutane (c) 1-Chloro-2-methylpropane (d) 2-Chloro-2-methylpropane * **Chapter:**

Haloalkanes and Haloarenes | **Confidence:** High

Q12. The geometry of the complex $[Ni(CO)_4]$ is: (a) Square planar (b) Tetrahedral (c) Octahedral (d) Trigonal bipyramidal * **Chapter:** Coordination Compounds | **Confidence:** High

Directions for Q13 to Q16: Select the correct answer from the codes (a), (b), (c), and (d). *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q13. Assertion (A): The order of a chemical reaction can be zero or a fractional number. **Reason (R):** The order of a reaction is determined strictly experimentally and is not necessarily equal to the sum of the stoichiometric coefficients in the balanced equation. * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q14. Assertion (A): Vitamin C cannot be stored in the human body. **Reason (R):** Vitamin C is a water-soluble vitamin and is readily excreted in urine. * **Chapter:** Biomolecules | **Confidence:** Very High

Q15. Assertion (A): The boiling point of a solvent increases upon the addition of a non-volatile solute. **Reason (R):** The vapour pressure of the solution becomes lower than that of the pure solvent at any given temperature. * **Chapter:** Solutions | **Confidence:** High

Q16. Assertion (A): The presence of a nitro ($-NO_2$) group at the ortho or para position increases the reactivity of haloarenes towards nucleophilic substitution. **Reason (R):** The nitro group stabilizes the carbanion intermediate formed during the reaction via resonance. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** High

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SECTION B: Very Short Answer Questions (2 Marks Each x 5 = 10 Marks)

Q17. Differentiate clearly between the *average rate* and the *instantaneous rate* of a chemical reaction. * **Chapter:** Chemical Kinetics | **Confidence:** High

Q18. Draw the structures of the optical isomers (enantiomers) of the complex ion $[Co(en)_2Cl_2]^+$. * **Chapter:** Coordination Compounds | **Confidence:** High

Q19. Outline the sequence of reagents to convert: (i) Nitrobenzene to Aniline (ii) Aniline to Bromobenzene * **Chapter:** Amines | **Confidence:** Very High

Q20. Account for the following observations: (i) The chromate ion (CrO_4^{2-}) is yellow while the dichromate ion ($Cr_2O_7^{2-}$) is orange, and they are interconvertible depending on the pH. (ii) The enthalpy of atomization of zinc is the lowest among the metals of the 3d transition series. * **Chapter:** d- and f-Block Elements | **Confidence:** High

Q21. State Raoult's law for a solution containing a non-volatile solute. Briefly explain why the vapour pressure of a solvent is lowered upon the addition of a non-volatile solute. * **Chapter:** Solutions | **Confidence:** Very High

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SECTION C: Short Answer Questions (3 Marks Each x 7 = 21 Marks)

Q22. The half-life ($t_{1/2}$) for a first-order reaction is exactly 30 minutes. Calculate the time required to complete 99% of the reaction. (Given: $\log 10 = 1$) * **Chapter:** Chemical Kinetics | **Confidence:** Very High

Q23. The conductivity of a $0.00241\,M$ solution of acetic acid is $7.896 \times 10^{-5}\,S\,cm^{-1}$. Calculate its molar conductivity (Λ_m). If the limiting molar conductivity (Λ_m°) for acetic acid is $390.5\,S\,cm^2\,mol^{-1}$, what is its acid dissociation constant (K_a)? * **Chapter:** Electrochemistry | **Confidence:** Very High

Q24. Answer the following: (i) Why is chloroform stored in dark-coloured, tightly closed bottles completely filled to the brim? (ii) What chemical products are formed when ethyl chloride is treated with *aqueous* KOH versus *alcoholic* KOH? Write the names of the respective major products. * **Chapter:** Haloalkanes and Haloarenes | **Confidence:** Very High

Q25. Provide the detailed, step-by-step mechanism for the acid-catalyzed dehydration of ethanol to yield ethene at 443 K. * **Chapter:** Alcohols, Phenols and Ethers | **Confidence:** Very High

Q26. Write the structures of the main organic products formed in the following reactions: (i) Benzene reacts with CO and HCl in the presence of anhydrous $AlCl_3$ (Gattermann-Koch reaction). (ii) Toluene reacts with CrO_2Cl_2 in CS_2 followed by acid hydrolysis (Etard reaction). (iii) Acetone reacts with hydrogen cyanide (HCN). * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

Q27. (i) Differentiate between essential and non-essential amino acids. Give one example of each. (ii) What is a peptide linkage, and how is it formed? * **Chapter:** Biomolecules | **Confidence:** High

Q28. Account for the following transition metal characteristics: (i) Transition metals and their compounds exhibit variable oxidation states. (ii) The standard reduction potential $E^\circ(M^{2+}/M)$ value for copper is positive (+0.34 V). (iii) Zinc ($Z = 30$) is not strictly considered a transition element. * **Chapter:** d- and f-Block Elements | **Confidence:** Very High

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SECTION D: Case Study-Based Questions (4 Marks Each x 2 = 8 Marks)

Q29. Case Study 1: Freezing Point Depression and its Applications The addition of a non-volatile solute to a solvent decreases its escaping tendency, lowering the vapour pressure. Consequently, the temperature at which the liquid solvent and solid solvent are in equilibrium—the freezing point—is depressed. This colligative property is directly proportional to the molality of the solution. The phenomenon has various practical applications, such as the use of ethylene glycol in car radiators to prevent the coolant from freezing during winter. **(i)** Define the cryoscopic constant (K_f). (1 Mark) **(ii)** Which aqueous solution will have a proportionally lower freezing point: $0.1\,M\,NaCl$ or $0.1\,M\,\text{Glucose}$? Give a reason. (1 Mark) **(iii)** Calculate the freezing point of an aqueous solution containing $10.5\,g$ of Magnesium bromide ($MgBr_2$) dissolved in $200\,g$ of water. (Assume complete dissociation of the salt. Molar mass of $MgBr_2 = 184\,g\,mol^{-1}$, $K_f \text{ for water} = 1.86\,K\,kg\,mol^{-1}$). (2 Marks) * **Chapter:** Solutions | **Confidence:** Very High

Q30. Case Study 2: Nucleophilic Addition to Carbonyl Compounds Aldehydes and ketones readily undergo nucleophilic addition reactions due to the highly polar nature of the carbonyl group. The carbonyl carbon carries a partial positive charge, making it an electrophilic center susceptible to attack by nucleophiles. However, steric hindrance (crowding by bulky alkyl groups) and electronic factors (+I effect of alkyl groups reducing the positive charge on carbon) strongly influence reactivity. **(i)** Based on the passage, state one primary reason why aldehydes are generally more reactive than ketones towards nucleophilic addition reactions. (1 Mark) **(ii)** Arrange the following compounds in increasing order of their reactivity towards nucleophilic addition: Formaldehyde, Acetone, Acetaldehyde. (1 Mark) **(iii)** Illustrate the general two-step mechanism of nucleophilic addition to a carbonyl group under basic conditions, showing the formation of the tetrahedral intermediate. (2 Marks) * **Chapter:** Aldehydes, Ketones and Carboxylic Acids | **Confidence:** High

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SECTION E: Long Answer Questions (5 Marks Each x 3 = 15 Marks)

Q31. (A) Electrochemistry: Cells & Faraday's Laws (i) Formulate the galvanic cell in which the following redox reaction takes place: $Zn_{(s)} + 2Ag^{+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + 2Ag_{(s)}$ (1 Mark) (ii) State Faraday's Second Law of Electrolysis. (1 Mark) (iii) A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using a steady current of 5.0 amperes for exactly 20 minutes. What mass of Nickel is deposited at the cathode? (Molar mass of Ni = $58.7\,g\,mol^{-1}$, $1F = 96500\,C\,mol^{-1}$). (3 Marks) * **OR (B)** (i) Define molar conductivity. Explain how molar conductivity varies with dilution for a strong electrolyte and sketch a graph representing this variation. (2 Marks) (ii) Calculate the standard cell potential (E°_{cell}) and the standard Gibbs free energy change (ΔG°) for the given cell reaction at 298 K: $2Cr_{(s)} + 3Cd^{2+}_{(aq)} \rightarrow 2Cr^{3+}_{(aq)} + 3Cd_{(s)}$ Given: $E^{\circ}_{Cr^{3+}/Cr} = -0.74\,V$, $E^{\circ}_{Cd^{2+}/Cd} = -0.40\,V$, and $1F = 96500\,C\,mol^{-1}$. (3 Marks) * **Chapter:** Electrochemistry | **Confidence:** Very High

Q32. (A) Organic Chemistry: Amines & Diazonium Salts Write the balanced chemical reactions involved in the following and state their specific reaction conditions: (i) Carbylamine reaction (Isocyanide test) (ii) Diazotization reaction of Aniline (iii) Coupling reaction of benzenediazonium chloride with phenol (iv) Reaction of aniline with excess bromine water (v) Protection of the amino group of aniline by acetylation (using acetic anhydride). * **OR (B)** (i) Give a distinct chemical test to distinguish between the following pairs of amines: a) Primary and secondary amines (using Hinsberg's reagent) b) Aliphatic primary amines and aromatic primary amines (ii) Account for the following reasoning questions: a) The pK_b of aniline is higher than that of methylamine. b) Ethylamine is completely soluble in water whereas aniline is nearly insoluble. c) Aniline does not undergo the Friedel-Crafts alkylation reaction. * **Chapter:** Amines | **Confidence:** Very High

Q33. (A) Coordination Compounds: Theories & Hybridization (i) Write the main postulates of Werner's theory of coordination compounds, specifically explaining primary and secondary valencies. (2 Marks) (ii) Using Valence Bond Theory (VBT), predict the hybridization, geometry, and magnetic behaviour of the hexafluoridocobaltate(III) complex ion, $[CoF_6]^{3-}$. (Given: atomic number of Co = 27, and fluoride is a weak field ligand). (3 Marks) * **OR (B)** (i) What is meant by the spectrochemical series? (1 Mark) (ii) Based on Crystal Field Theory (CFT), write the electronic configuration of a d^4 central metal ion in terms of t_{2g} and e_g orbitals in an octahedral field when: a) $\Delta_o < P$ (Weak field ligand) b) $\Delta_o > P$ (Strong field ligand) (2 Marks) (iii) Write the systematic IUPAC names for the following coordination complexes: a) $K_3[Fe(C_2O_4)_3]$ b) $[Pt(NH_3)_2Cl(NO_2)]$ (2 Marks) * **Chapter:** Coordination Compounds | **Confidence:** Very High

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